



Enhancing Data and Knowledge Discoverability at NASA GES-DISC through Knowledge Graphs and Language Models

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Abstract

The Goddard Earth Sciences Data and Information Services Center (GES-DISC) at NASA curates an extensive collection of over 1500 satellite datasets, covering diverse Earth Science Focus Areas including Atmospheric Composition, Water & Energy Cycles, Climate Variability, and the Carbon Cycle & Ecosystem. Despite meticulous curation and continuous enhancement of information services, the existing metadata might not suffice to address the intricate user needs for dataset search, discovery, and utilization, particularly given the vast unstructured sources associated with satellite datasets. This project introduces an innovative framework designed at GES-DISC that employs cutting-edge, open-source technologies to revolutionize the way users interact with data and extract information. Built on the pillars of Language Models (LMs), Graph Databases, Vector Databases, and intuitive user interface tools like Streamlit, the framework offers an end-to-end solution for dataset queries and scientific inquiries.

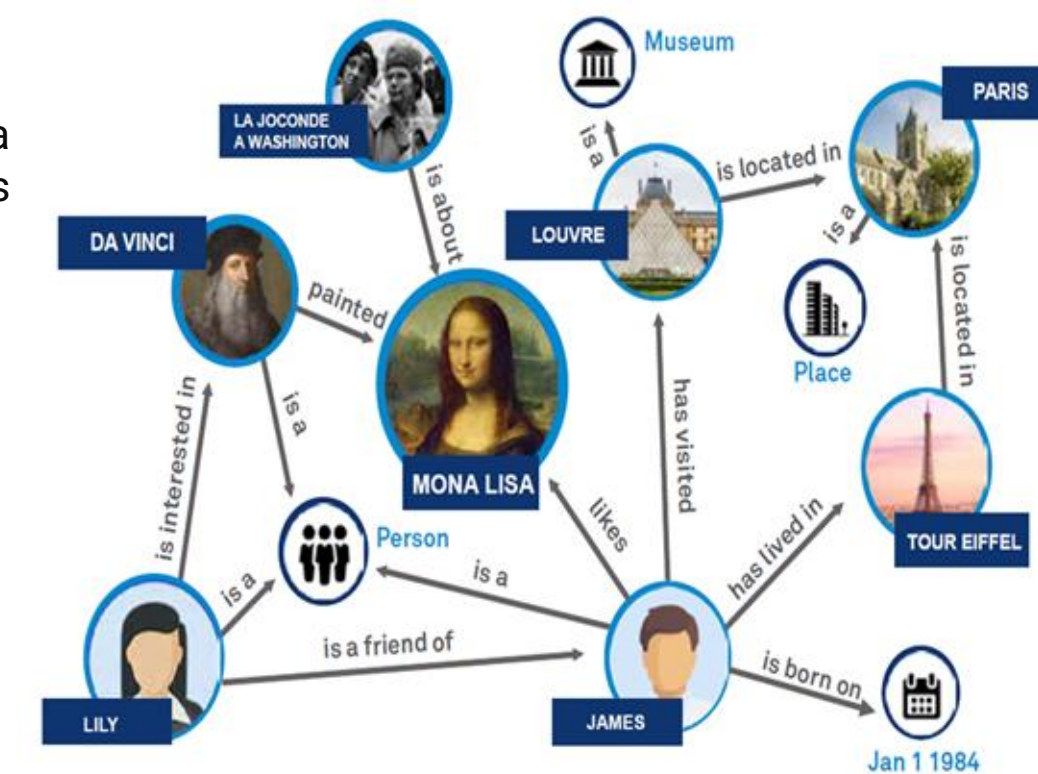
Introduction

In the heart of this transformative approach is the integration of open-source resources, a crucial step promoting open science. LMs are utilized to derive insightful attributes from unstructured data, enhancing the existing metadata, while Graph Databases allow for an intuitive and relational organization of this data. Vector Databases enable high-speed similarity searches, bringing forth the most relevant datasets in response to user queries. This entire system is made accessible through Streamlit, an open-source, user-friendly interface tool.

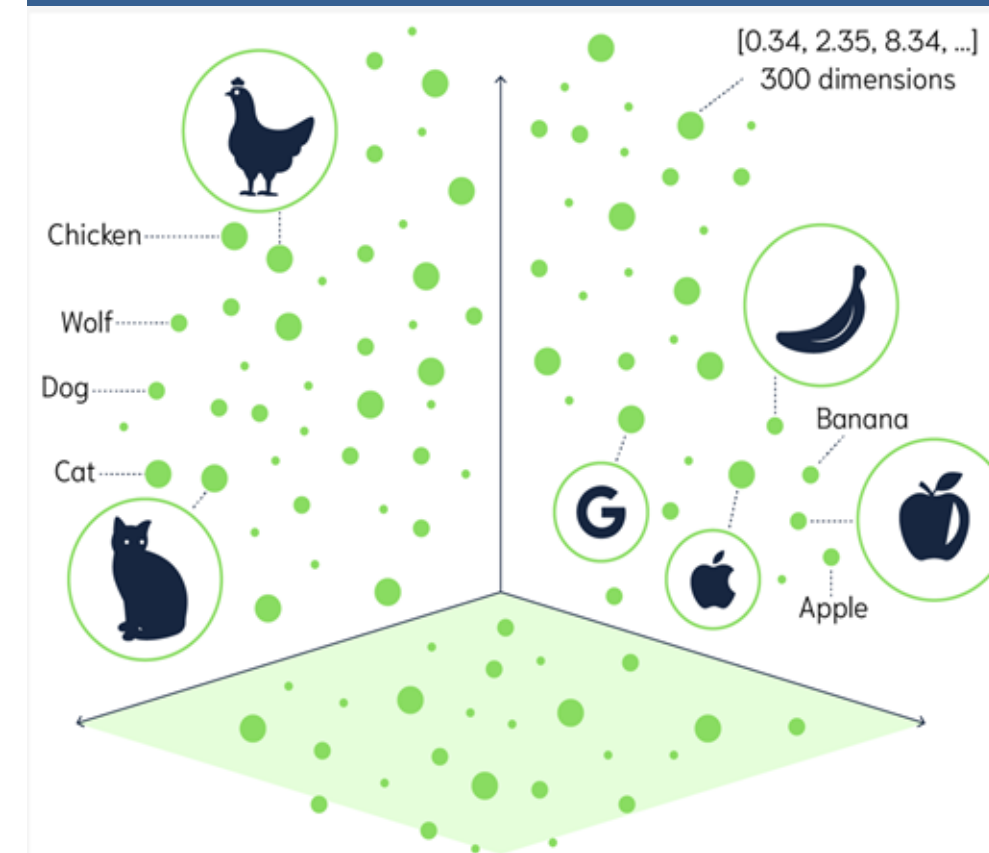
This robust, comprehensive approach not only bolsters the discoverability and usability of GES-DISC's vast data resources but also aligns with the NASA GES-DISC's primary goal of optimally serving the Earth science community's data and information needs. By leveraging open-source tools, we are encouraging collaboration, transparency, and accessibility in scientific exploration. The potential of this framework extends beyond NASA, offering a scalable and adaptable model for data management and user interaction that can serve various scientific disciplines and data-intensive applications.

1. Knowledge Graphs

Knowledge graphs are structured representations of knowledge that capture relationships between entities in a domain. They organize information in a graph-like structure, where nodes represent entities (such as people, places, or concepts) and edges represent relationships between them. Knowledge graphs are powerful tools for organizing, connecting, and analyzing vast amounts of information from various sources, such as text documents, databases, and the web. They enable efficient querying, reasoning, and inference, facilitating tasks like semantic search, recommendation systems, and data integration. Knowledge graphs play a crucial role in advancing artificial intelligence by providing a foundation for knowledge representation and semantic understanding, ultimately enabling smarter and more context-aware applications.



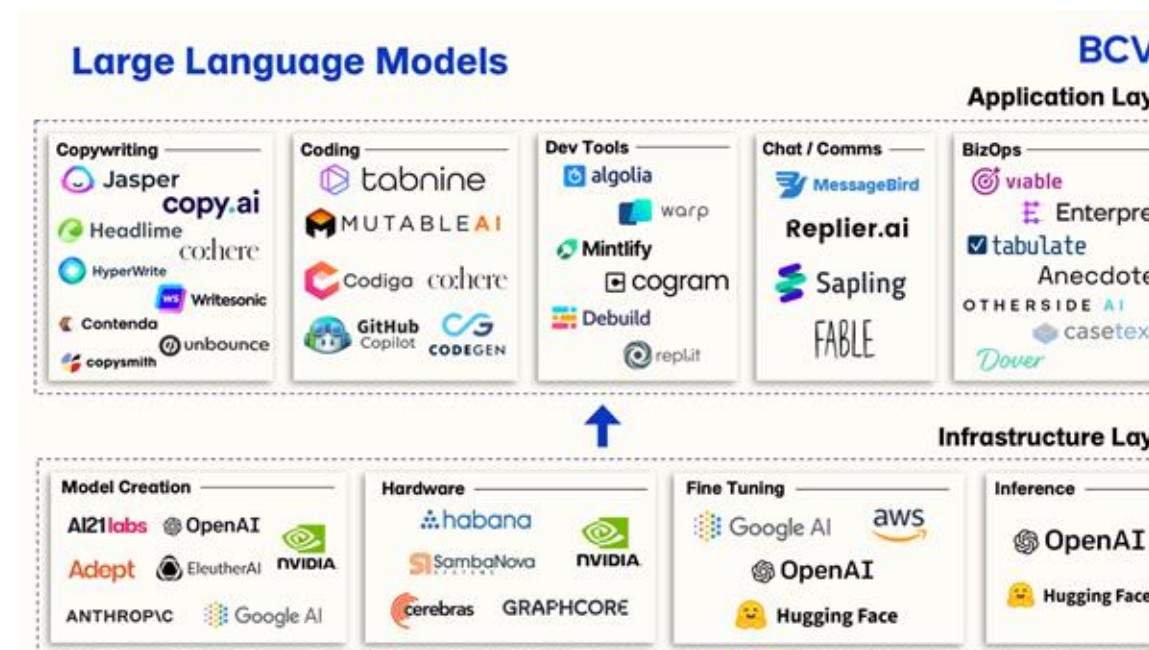
2. Vector Database



A vector database is a specialized database designed to efficiently store and manage vector data, where each vector represents a data point in a multi-dimensional space. Unlike traditional relational databases that organize data in tables, vector databases handle collections of vectors, making them ideal for applications requiring spatial or geometric data representation. Commonly used in fields such as geographic information systems (GIS) and machine learning, vector databases excel at performing complex geometric operations and similarity searches. By providing optimized indexing and querying mechanisms for high-dimensional data, vector databases enable fast and scalable processing, making them invaluable tools for analyzing and extracting insights from diverse datasets.

3. Language Models

Language models are advanced AI systems trained on vast amounts of text data to understand and generate human-like language. They utilize deep learning techniques to learn patterns, structures, and relationships within language, enabling them to perform various natural language processing tasks. These tasks include text generation, translation, summarization, sentiment analysis, and more. Language models have revolutionized many aspects of human-computer interaction, powering virtual assistants, chatbots, language translation services, and content generation tools. With ongoing advancements in AI research, language models continue to evolve, offering increasingly sophisticated capabilities in understanding and generating human language.



Implementation

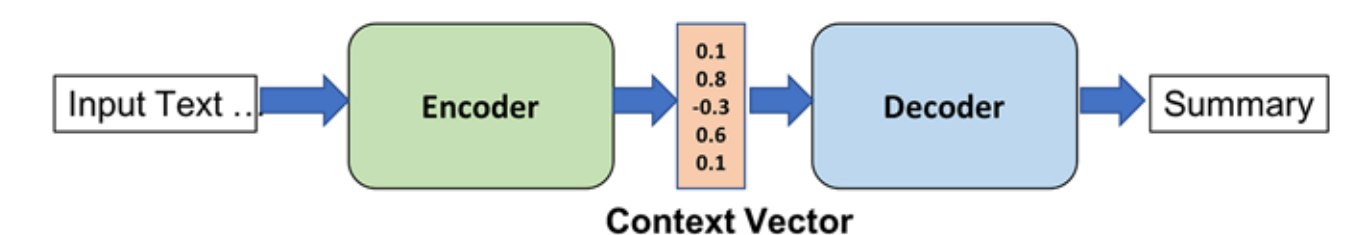
To complete the goal of this project I began by cleaning the data. I was given 530 JSON files representative of metadata surrounding different datasets stored by NASA, and 1960 PDFs in which the datasets were referenced. To clean the data I created a regular expression to extract the pertinent information from each JSON and PDF.

```
keywords = ["Introduction", "Description", "Technical Summary", "Extended User Guide"]
for keyword in keywords:
    match = re.search(rf'{keyword}\s+(?=[\w\d])', pdf_content, re.IGNORECASE)
```

To merge the cleaned JSON and PDF files to create a cohesive search engine for Earth Science researchers I began by creating a graph database in Neo4j. Through this visual representation of a knowledge graph I was able to successfully identify the different hierarchies within the data. From this portion of the project I moved into developing a vector database in Weaviate. The vector database is the "brain" of the application and allows the computer to identify the connections between each entity within the knowledge graph. Once the vector database was created I integrated it as the backend of a Streamlit application. Streamlit is an open-source application I utilized to create the front-end of my application. Streamlit allows me to create an application in which the user could ask a question about the Earth Science datasets or PDFs and the results would be a list of PDFs pertaining to the question along with the associated datasets.



After the successful integration of the vector database and Streamlit moved to the final step, a chatbot integration. For this chatbot function I utilized an encoder-decoder transformer. This special type of transformer allowed the summaries of each PDF to be better contextualized for better results to questions posed by the users.



Conclusion

This project integrates knowledge graphs, vector databases, transformers, and open-source applications to enhance data discoverability of Earth Science datasets and PDFs within NASA databases. Future improvements include integrating unstructured data to further enhance discoverability for research. An area for enhancement is developing a unique transformer specifically trained on summarized data. Due to time constraints, a pre-trained transformer from Hugging Face was used as a proof of concept. However, a transformer trained on summaries and dataset metadata is essential for a fully functioning application. Additionally, improving the application's visual design to align more closely with NASA's traditional theme is necessary. Overall, this project combines machine learning, web development, and data science to create an advanced application for improved NASA research discoverability.